Thin perfect matchings [HOG26]: Lean formalisation

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. A. Haqi, S. Oveis Gharan. *On Thin Perfect Matchings up to Polylogarithmic Factors.* arXiv:2606.01330v2 (2026) [HOG26].

The library `ThinMatching/` (independent of `GsimOT`) formalises [HOG26]: Lemma 1.2, Theorem 1.3 (support-constrained) and Theorem 1.4 (relaxed, bipartite and general) with all lemmas of Sections 3 and 4. Theorem 1.4 is the cited Theorem 35 of the main paper (p. 32). A proof overview is in `ThinMatching/HOG26_overview.pdf`.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `ThinMatching.theorem14_bip`: assuming [RS14] (`RackeShahTheorem`), there is an absolute $C > 0$ such that for every bipartite fractional perfect matching x on $L \cup R$, $|L| = |R| = n$, some perfect matching $M \subseteq L \times R$ satisfies $|M \cap \delta(S)| \leq C \ln(2n) \ln\ln(4n)\, x(\delta(S))$ for all $S \subseteq L \cup R$. `ThinMatching.theorem13_support`: assuming Lemma 3.6 (`Lemma36`), every $x \in \mathrm{PM}(G)$ has a $C|V| \ln(2|V|)$-thin perfect matching inside $\mathrm{supp}(x)$.

Correspondence

Paper	Page	Statement	Lean declaration(s) and location

Sec. 1, Sec. 2	1, 4	Weighted graph $G = (V, E, x)$, $x(\delta(S))$, perfect matchings, $\lvert M \cap \delta(S)\rvert$, α-thin, $M \subseteq \mathrm{supp}(x)$	`ThinMatching.WGraph`
			ThinMatching/Basic.lean:30
			`ThinMatching.cutWeight`
			ThinMatching/Basic.lean:41
			`ThinMatching.PerfectMatching`
			ThinMatching/Basic.lean:44
			`ThinMatching.crossCount`
			ThinMatching/Basic.lean:50
			`ThinMatching.IsThin`
			ThinMatching/Basic.lean:53
			`ThinMatching.InSupport`
			ThinMatching/Basic.lean:57

Sec. 1, Thm. 2.5	1, 5	$x \in \mathrm{PM}(G)$: convex combination of perfect matchings; bipartite version (degree constraints)	`ThinMatching.IsFracPM`
			ThinMatching/Basic.lean:60
			`ThinMatching.IsFracPMBip`
			ThinMatching/Basic.lean:84
			`ThinMatching.bipGraph`
			ThinMatching/Basic.lean:90

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Thm. 1.4 (bipartite)	1, 2	Bipartite cut weight, crossing count and thinness for $M = \{(l, \pi l)\}$	`ThinMatching.bipCutWeight`
			ThinMatching/Basic.lean:71
			`ThinMatching.bipCrossCount`
			ThinMatching/Basic.lean:75
			`ThinMatching.IsThinBip`
			ThinMatching/Basic.lean:79

Lemma 1.2	1	Every perfect matching in $\mathrm{supp}(x)$ of the graph of Fig. 1 ($2n+2$ vertices) is at best $\frac{n}{2}$-thin	`ThinMatching.lemma12`
			ThinMatching/LowerBound.lean:200
			`ThinMatching.lemma12_not_thin`
			ThinMatching/LowerBound.lean:208

Fig. 1	1	The graph: solid edges $1-\frac{1}{n}$, dotted edges $\frac{1}{n}$; x as the average of n perfect matchings	`ThinMatching.LowerBound.fig1`
			ThinMatching/LowerBound.lean:43
			`ThinMatching.LowerBound.Mk`
			ThinMatching/LowerBound.lean:74
			`ThinMatching.LowerBound.fig1_isFracPM`
			ThinMatching/LowerBound.lean:83
			`ThinMatching.LowerBound.fig1_bad_cut`
			ThinMatching/LowerBound.lean:135

Theorem 1.3	2	Support-constrained $O(n \log n)$-thin perfect matching (assuming Lemma 3.6)	`ThinMatching.theorem13_support` ThinMatching/SupportConstrained.lean:212

Def. 3.1, Lemma 3.2	5	Trimming $\tilde{x}$; removed weight $\leq (\lvert V\rvert - 1)\varepsilon$; components ε-edge-connected	`ThinMatching.trimGraph`
			ThinMatching/SupportConstrained/Trimming.lean:45
			`ThinMatching.IsTrimmed`
			ThinMatching/SupportConstrained/Trimming.lean:52
			`ThinMatching.exists_split`
			ThinMatching/SupportConstrained/Trimming.lean:84
			`ThinMatching.trimming_exists`
			ThinMatching/SupportConstrained/Trimming.lean:156
			`ThinMatching.lemma32`
			ThinMatching/SupportConstrained/Trimming.lean:213

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Algorithm 1	6	Independent sampling of α copies of each edge with probability $\tilde{x}_e$ (finite product space)	ThinMatching.bernW ThinMatching/SupportConstrained/Prob.lean:27 ThinMatching.prob ThinMatching/SupportConstrained/Prob.lean:31 ThinMatching.sampleProb ThinMatching/SupportConstrained/Sampling.lean:67 ThinMatching.sampleCut ThinMatching/SupportConstrained/Sampling.lean:81 ThinMatching.sampleGraph ThinMatching/SupportConstrained/Sampling.lean:85
Thm. 2.4, Def. 3.3	5	Tutte's theorem (Mathlib) and Tutte witnesses	ThinMatching.IsTutteWitness ThinMatching/SupportConstrained/TutteWitness.lean:50 ThinMatching.witnessF ThinMatching/SupportConstrained/TutteWitness.lean:62 ThinMatching.exists_witness_of_no_pm ThinMatching/SupportConstrained/TutteWitness.lean:170
Lemma 3.4	6	At most $(\lvert V\rvert+1)^{\lvert V\rvert}$ Tutte witnesses	ThinMatching.card_witnesses_le ThinMatching/SupportConstrained/TutteWitness.lean:86
Lemma 3.5, Eq. (1)	6	The sample contains a perfect matching w.h.p.; $x(F)\geq 1$ (odd-set argument)	ThinMatching.lemma35 ThinMatching/SupportConstrained/Lemma35.lean:62 ThinMatching.one_le_witnessF_weight ThinMatching/SupportConstrained/TutteWitness.lean:134 ThinMatching.witness_bound_le ThinMatching/SupportConstrained/Lemma35.lean:122
Lemma 3.6	7	Sampling preserves cuts up to $O(\alpha)$ w.h.p. (proof omitted in [HOG26]) — **explicit hypothesis**	ThinMatching.Lemma36 ThinMatching/SupportConstrained/Sampling.lean:120
Proof of Thm. 1.3	7	Union over components; thinness for all cuts; $M\subseteq Y\subseteq \mathrm{supp}(\tilde{x})\subseteq \mathrm{supp}(x)$	ThinMatching.sum_cut_eq_sum_blocks ThinMatching/SupportConstrained.lean:72 ThinMatching.crossCount_le_of_good ThinMatching/SupportConstrained.lean:119 ThinMatching.prob_bad_blocks_le ThinMatching/SupportConstrained.lean:174

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Def. 2.1	4	Weighted tree with Steiner nodes (rooted, parent function); $y(\delta_{\mathcal{T}}(\mathcal{S}))$; $\lambda_{\mathcal{T}}(S)$; q-tree-cut-sparsifier	`ThinMatching.RootedTree`
			ThinMatching/TreeSparsifier.lean:29
			`ThinMatching.subtree`
			ThinMatching/TreeSparsifier.lean:39
			`ThinMatching.treeCutWeight`
			ThinMatching/TreeSparsifier.lean:44
			`ThinMatching.lambdaT`
			ThinMatching/TreeSparsifier.lean:50
			`ThinMatching.IsTreeCutSparsifier`
			ThinMatching/TreeSparsifier.lean:55

Thm. 2.2 [RS14]	4	Every graph has an $O(\log n \log\log n)$-tree-cut-sparsifier — **explicit hypothesis**	`ThinMatching.RackeShahTheorem` ThinMatching/TreeSparsifier.lean:63

Defs. 4.1, 4.2, 4.4	7–8	Tree cuts, load of a matching on a tree edge, discrepancy Δ	`ThinMatching.load`
			ThinMatching/Load.lean:32
			`ThinMatching.loadBip`
			ThinMatching/Load.lean:46
			`ThinMatching.discBip`
			ThinMatching/Load.lean:50
			`ThinMatching.discNonBip`
			ThinMatching/Load.lean:36

Proposition 4.3	7	$\mathrm{load}_M(f) \le \beta\, x(\delta(V_r(f)))$ for all tree edges $\Rightarrow$ M is βq-thin	`ThinMatching.prop43`
			ThinMatching/LoadVsCut.lean:170
			`ThinMatching.LoadVsCut.exists_cut_edge`
			ThinMatching/LoadVsCut.lean:107
			`ThinMatching.LoadVsCut.treeCutWeight_subtree`
			ThinMatching/LoadVsCut.lean:148

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Lemma 4.5	8	$x(\delta(S)) \geq \Delta_{\text{bip}}(S)$; $x(\delta(S)) \geq 1$ for $\lvert S\rvert$ odd	`ThinMatching.lemma45_bip`
			`ThinMatching/Discrepancy.lean:21`
			`ThinMatching.lemma45`
			`ThinMatching/Discrepancy.lean:81`
			`ThinMatching.one_le_crossCount_of_odd`
			`ThinMatching/Discrepancy.lean:43`
			`ThinMatching.cutWeight_eq_sum_crossCount`
			`ThinMatching/Discrepancy.lean:58`

Lemma 4.6, Algorithm 2	9–10	Tree pairing: a perfect matching with $\text{load}_M(f) = \Delta(V_r(f))$ for every tree edge	`ThinMatching.lemma46_bip`
			`ThinMatching/TreePairing.lean:135`
			`ThinMatching.lemma46`
			`ThinMatching/TreePairing.lean:212`
			`ThinMatching.subtree_laminar`
			`ThinMatching/TreePairing.lean:56`
			`ThinMatching.subtree_ssubset_of_separates`
			`ThinMatching/TreePairing.lean:70`

Theorem 1.4 (bipartite)	2	An $O(\log n \log\log n)$-thin perfect matching $M \subseteq L \times R$ (assuming [RS14])	`ThinMatching.theorem14_bip`
			`ThinMatching/Theorem14.lean:135`
			`ThinMatching.Theorem14.bipPM`
			`ThinMatching/Theorem14.lean:30`
			`ThinMatching.Theorem14.cutWeight_bipGraph`
			`ThinMatching/Theorem14.lean:64`
			`ThinMatching.Theorem14.log_mul_loglog_le`
			`ThinMatching/Theorem14.lean:96`

Theorem 1.4 (general)	2	An $O(\log n \log\log n)$-thin perfect matching $M \subseteq \binom{V}{2}$ for $x \in \text{PM}(G)$ (assuming [RS14])	`ThinMatching.theorem14` `ThinMatching/Theorem14.lean:180`

Main paper, Thm. 35, Eq. (23)	–	Bridge to `GsimOT`: Theorem 35 (p. 32, thinness Eq. (23)) and Theorem 11 of the main paper assuming only [RS14]	`GsimOT.thinPermutationTheorem_of_rackeShah`
			`Theorem13_CutUpperBound/FromThinMatching.lean:18`
			`GsimOT.theorem13_of_rackeShah`
			`Theorem13_CutUpperBound/FromThinMatching.lean:28`
			`GsimOT.theorem13_power_of_rackeShah`
			`Theorem13_CutUpperBound/FromThinMatching.lean:38`

Notes on the formalisation

1. **Independence.** The library `ThinMatching` imports only Mathlib and does not depend on the `GsimOT` formalisation. It has its own definitions of weighted graphs, cuts, perfect matchings, thinness and trees. The only connection is the bridge file `GsimOT/Theorem13_CutUpperBound/FromThinMatching.lean`, which derives the main paper's Theorem 35 from the bipartite case of Theorem 1.4. The main paper's α-thinness, Eq. (23) (cuts $(X \cup Y, \overline{X} \cup \overline{Y})$ of $K_{n,n}$, all $X, Y \subseteq [n]$), is exactly `IsThinBip` with $S_L = X$, $S_R = Y$, and its constant $C_{\text{thin}} \ln(2n) \ln\ln(4n)$ is the one of `theorem14_bip` (same C).

2. **Cited results.** Theorem 2.2 (Räcke–Shah [RS14], tree-cut sparsifiers) is used but not proved in [HOG26]. It is the explicit hypothesis `RackeShahTheorem`, with the bound $C \ln(2n) \ln\ln(4n)$, which is positive for all $n \geq 1$. Lemma 3.6 is stated in [HOG26] with its proof omitted ("a simple Chernoff bound and a cut-counting argument"). It is the explicit hypothesis `Lemma36`, formulated in the finite sampling model of Algorithm 1, for any $\alpha \geq cN \ln(2N)$ with $N \geq |W|$, with failure probability $\leq 1/(4N)$. This is what the omitted argument (Karger's cut counting + Chernoff) yields. No Lean `axiom` is introduced; Theorems 1.3 and 1.4 take these hypotheses as arguments.

3. **Trees.** A weighted tree with Steiner nodes is represented as a rooted tree on a finite type via a parent function, with $V \hookrightarrow \mathcal{V}$; the tree edges are $\{v, \text{parent}(v)\}$. Every tree can be rooted, so this is no loss of generality. The tree path between two vertices uses the edge $\{v, \text{parent}(v)\}$ iff exactly one of them lies in the subtree of v; this is how loads are defined. Discrepancies are taken on the subtree side $V_s(f)$, which equals the root side $V_r(f)$ of the paper since $|L| = |R|$ resp. $|V|$ is even.

4. **Lemma 4.6 (Algorithm 2).** Instead of executing the greedy vertex by vertex, the proof matches, one pair at a time, a matchable pair inside a *minimal* subtree that contains one. This is the first pair the greedy forms. The paper's main observation (no proper subtree of it contains a matchable pair) shows that load and discrepancy change in the same way, which gives load $= \Delta$ by induction.

5. **Proposition 4.3.** The charging step uses a lemma (`exists_cut_edge`) finding, for $a \in \mathcal{S}$, $b \notin \mathcal{S}$, a tree edge cut by $\mathcal{S}$ that separates a and b, by induction on the depths instead of via the explicit path. Lemma 4.5 in the general case is derived from the convex combination of perfect matchings (each crosses an odd set), the easy direction of Theorem 2.5.

6. **Theorem 1.3.** The paper argues per component. The formalisation also checks cuts through several components: sampled edges lie inside blocks, so the per-component bounds add up. Further choices: a union bound over at most $|V|$ components, $\varepsilon = 1/(4|V|)$ (the paper's $1/(8n)$ for $|V| = 2n$), and $x(F) \geq 1$ (slightly stronger than the paper's route). Trimming is modelled by block labellings, and each step adds one block.

7. **Not formalised.** Corollary 1.6 (metric distortion) depends on [ACR23, Prop. 3] and on mechanism definitions outside the mathematical core of the paper. Theorem 2.3 (algorithmic sparsifiers) is not used.

8. Lemma 1.2 is proved for every $n \geq 1$; as in the paper, one crossing matching edge is counted (ratio $\geq n/2$).